STUDY MODULE 08

Introduction to Generative AI

From Recognizing Patterns to Creating Original Content

We live in an era where computers can write essays, compose music, generate hyper-realistic artwork, and code software programs in seconds. This exciting technology is called **Generative AI**.

1. What Makes Generative AI Different?

Traditional AI and machine learning models are designed to *analyze* data and make decisions or predictions (such as spotting spam or diagnosing an illness). Generative AI goes a step further: it takes what it has learned and **creates brand-new, original content** that has never existed before.

> **The Core Analogy:** If a traditional AI model is like an expert art critic who can look at a painting and tell you who painted it, Generative AI is like an artist who can paint a brand-new masterpiece in that exact same style upon request.

2. How Generative AI Works

Generative AI is built on top of deep learning neural networks trained on massive amounts of internet data. When you type a prompt (a text instruction), the model does not look up a pre-written answer. Instead, it predicts the most likely next word, pixel, or note step-by-step based on patterns it memorized during training.

3. Different Forms of Generative AI

Generative models come in several different modalities depending on what type of output they produce:

Modality	What It Creates	Popular Examples
Text Models	Essays, computer code, summaries, and conversational responses.	ChatGPT, Claude, Gemini
Image Models	Digital paintings, photo-realistic pictures, and graphic designs from text prompts.	Midjourney, DALL-E, Stable Diffusion
Audio & Video	Realistic voice clones, musical compositions, and synthetic video clips.	Sora, ElevenLabs, Suno

4. Important Limitations and Risks

While generative AI is powerful, students should keep its limitations in mind:

- **Hallucinations:** Models sometimes state completely false information with absolute confidence.
- **Bias:** Because they learn from human data, they can pick up and repeat human prejudices.

> **Summary:** Generative AI transforms human-computer interaction by shifting computers from tools that follow rigid rules to creative partners that help generate ideas.

5. Review Checklist

- How does generative AI differ from traditional analytical AI?
- What is a prompt, and how does the model use it to generate a response?
- What is a "hallucination" in the context of generative text models?